TECHNICAL ARCHITECTURE & DATA ENGINEERING

EduVision: End-to-End Higher Education Analytics Platform

Scalable ETL Pipeline, 4-Fact Star Schema Architecture, and Tableau Executive Visual Analytics.

Python & Pandas ETL | 4-Fact Star Schema | Tableau Public / Server

SYSTEM PIPELINE

End-to-End Data Processing & Delivery Flow

1

Kaggle Extraction

Raw global university rankings and demographics datasets ingested.

2

Python / Jupyter ETL

Data cleaning, null imputation, key generation, and metric ratio calculations.

3

4-Fact Data Modeling

Structured into 2 conformed dimensions and 4 specialized fact tables.

4

Tableau Dashboards

4 targeted 1366×768 dashboards with context filtering and actions.

5

Deployment

Published to Tableau Public for interactive executive decision-making.

DATA INGESTION & TRANSFORMATION

Phase 1: Python Data Processing (Jupyter)

Multi-Source Raw Ingestion

Downloaded multi-year global rankings, research impact metrics, and country demographic data from Kaggle.

Data Standardization & Cleaning

Harmonized country names, handled missing values, and generated unique Surrogate Keys (`university_id` , `country_id`).

Pre-Aggregations

Computed complex ratios in Pandas prior to loading to maximize downstream Tableau dashboard rendering performance.

```
# Jupyter Notebook - Data Processing snippet

import pandas as pd
import numpy as np

# Surrogate Key Generation & Metric Calculations
df['intl_student_pct'] = (df['intl_students'] /
df['total_students']) * 100
df['student_staff_ratio'] = df['total_students'] / df['total_staff']

# Export Fact & Dimension tables
fact_overview.to_csv('fact_university_overview.csv', index=False)
fact_research.to_csv('fact_research_analytics.csv', index=False)
```

DATA WAREHOUSE MODELING

Phase 2: Star Schema Architecture (4 Fact Tables)

CONFORMED DIMENSION

dim_country

PK: country_id | Country Name, Region, Global Indicators

CONFORMED DIMENSION

dim_university

PK: university_id | FK: country_id | University Name, Location



FACT TABLE 1

fact_university_overview

Annual rank metrics, overall scores, global position benchmarks.

FACT TABLE 2

fact_research_analytics

Research outputs, citation impact scores, academic reputation indicators.

FACT TABLE 3

fact_student_analytics

Total enrollment, international student counts, faculty-to-student ratios.

FACT TABLE 4

fact_country_comparison

Macro country benchmarks, aggregate higher education indicators.

TABLEAU ARCHITECTURE

Phase 3 & 4: Interactivity & Filter Context Rules

Filter Hierarchy Rules

- **Year Filter:** Set to Global scope across all 4 dashboards.
- **Country Filter:** Cascading dropdown configured with *"Only Relevant Values"*.
- **University Filter:** Scoped to detail sheets while excluding Top-10 rank leaderboards to prevent collapse.

Context Filters & Actions

- **Context Filters (Grey Pills):** Added geographic filters to Context so Top-10 limits evaluate *after* country selections.
- **Inter-Dashboard Actions:** Configured `university_id` and `country_id` actions to seamlessly jump between Overview, Research, Student, and Country sheets.

TECHNICAL ARCHITECTURE MAP

Data Warehouse Schema & Source Details

Data Integration Specs

- **ETL Runtime:** Python 3.10+ in Jupyter Notebook.
- **Key Mapping:** Surrogate Keys linked via 1–to–Many relationships.
- **Optimization:** Star Schema reduces redundant string data storage by ~65% over flat tables.
- **Canvas Standard:** Fixed 1366×768 layout across all 4 dashboards.

Interactive Analytics Location

Published live on Tableau Public for stakeholder review and interactive exploration:

https://public.tableau.com/views/EduViz/ResearchAnalytics

Source: public.tableau.com

DEPLOYMENT READY

EduVision Analytics System Complete

Published on Tableau Server / Public for executive decision-making and higher education insights.

View Interactive Tableau Dashboard

Source: public.tableau.com